DHCP Snooping + Dynamic ARP Inspection

Mitigation of Rogue DHCP and ARP Spoofing Attacks using DHCP Snooping and Dynamic ARP Inspection (DAI)

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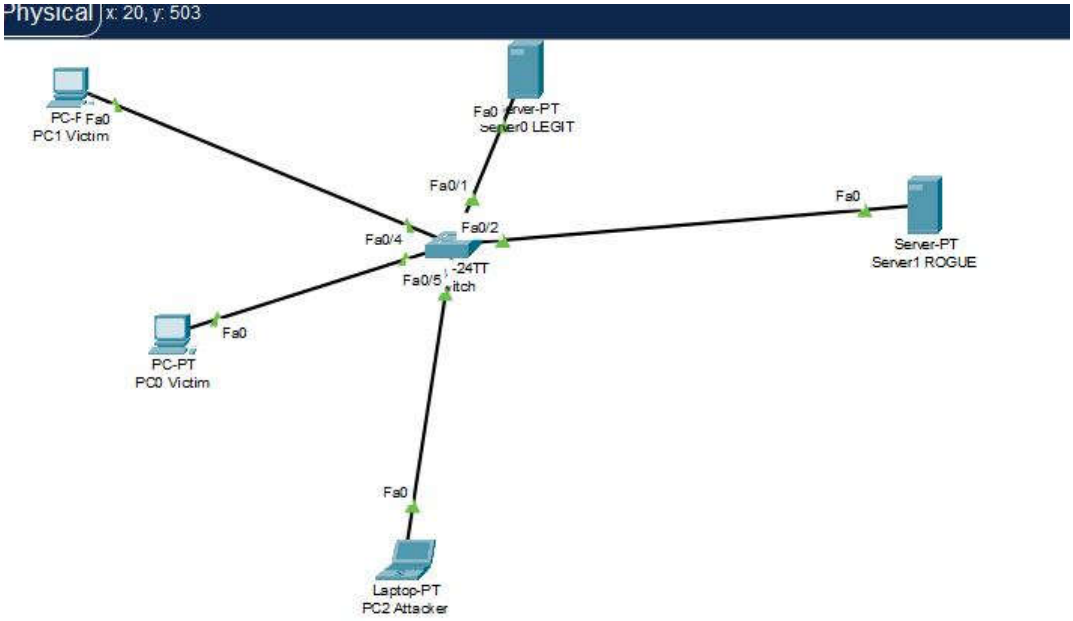
1. Objective

To demonstrate how a Layer 2 switch can be secured against two common LAN attacks:

- **Rogue DHCP Server Attack:** An unauthorized DHCP server giving wrong IPs / acting as Man-in-the-Middle.
- **ARP Spoofing / ARP Poisoning:** An attacker faking ARP replies to intercept traffic.

And to implement Cisco security features to block both.

2. Topology



All devices are in VLAN 1, Network 192.168.10.0/24.

3. IP Addressing Table

Device	Interface	IP Address	Role
Server0 Legit DHCP	F0	192.168.10.1 /24	Legitimate DHCP Server
Server1 Rogue DHCP	F0	192.168.10.99 /24	Rogue / Attacker DHCP
PC0	F0	DHCP	Victim 1

Device	Interface	IP Address	Role
PC1	F0	DHCP	Victim 2
PC2 Attacker	F0	192.168.10.20 static	ARP Spoof Attacker

4. Theory

a) DHCP Snooping

A switch security feature that creates a binding table of legitimate IP-to-MAC mappings from trusted DHCP responses. It filters all DHCP messages on untrusted ports. Only the port marked `trust` can send DHCP OFFER/ACK.

b) Dynamic ARP Inspection (DAI)

Uses the DHCP Snooping binding database to validate ARP packets. If an ARP reply has an IP/MAC that does not match the binding table, it is dropped. It also has rate limiting to prevent ARP flooding DoS.

5. Configuration Steps

Step A - Configure Servers

- **Server0 > Services > DHCP:** Pool 192.168.10.0/24, Gateway 192.168.10.1, Start IP 192.168.10.10, ON.
- **Server1 > Services > DHCP:** Pool 10.0.0.0/24, Start IP 10.0.0.10, ON.

Step B - Switch Hardening (Core of Lab)

On SW1 CLI:

```
enable
configure terminal
ip dhcp snooping
ip dhcp snooping vlan 1
no ip dhcp snooping information option

! ONLY LEGIT SERVER PORT IS TRUSTED
interface fastEthernet0/1
 description LEGIT_DHCP_TRUSTED
 ip dhcp snooping trust
 ip arp inspection trust
exit

! ALL OTHER PORTS ARE UNTRUSTED
interface range fastEthernet0/2 - 5
 description UNTRUSTED - Clients + Rogue
 no ip dhcp snooping trust
 no ip arp inspection trust
 ip arp inspection limit rate 15
exit

! ENABLE DAI
ip arp inspection vlan 1
ip arp inspection validate src-mac dst-mac ip
exit
write memory
```

6. Verification and Demonstration

Demo 1: Rogue DHCP Blocked

1. Both DHCP servers ON.
2. On PC0: `ipconfig /release` -> `ipconfig /renew`
3. Result: Gets `192.168.10.10` from Legit only.
4. Commands:

```
show ip dhcp snooping
show ip dhcp snooping binding
show ip dhcp snooping statistics
```

```
Switch>
Switch>
Switch>show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
1
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
FastEthernet0/1            yes        unlimited
FastEthernet0/3            no         unlimited
Switch>
Switch>
Switch>
```

```
Switch>
Switch>show ip dhcp snooping binding
MacAddress          IpAddress        Lease(sec)  Type           VLAN
Interface
------------------  ---------------  ----------  -------------  ----
------------------
00:01:96:08:43:E2   192.168.10.10    86400       dhcp-snooping  1
FastEthernet0/3
Total number of bindings: 1
Switch>
```

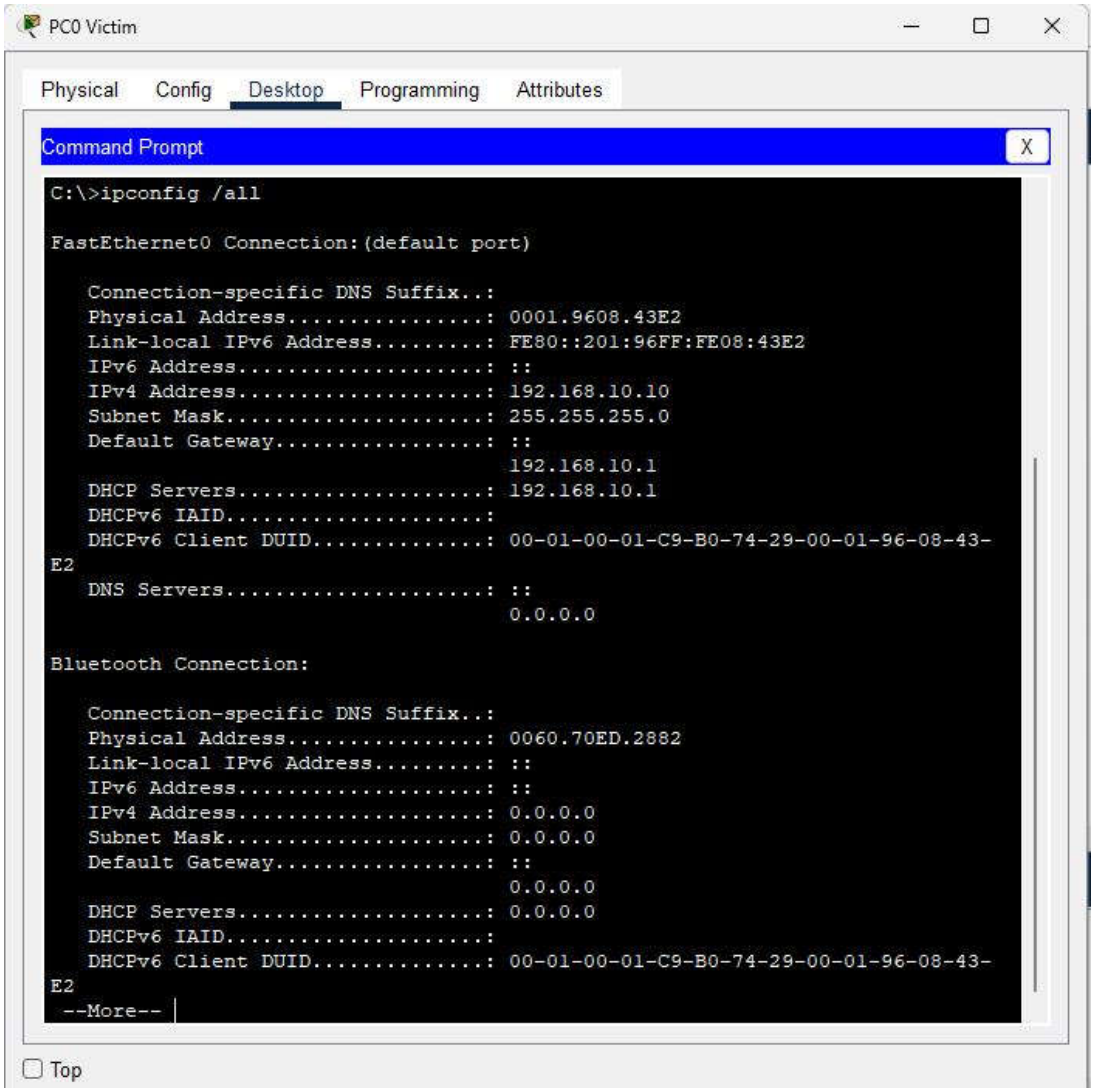
```
Switch#show ip arp inspection statistics vlan 1

 Vlan      Forwarded        Dropped     DHCP Drops     ACL Drops
 ----      ---------        -------     ----------     ---------
    1              0              0              0             0

 Vlan    DHCP Permits    ACL Permits   Source MAC Failures
 ----    ------------    -----------   -------------------
    1              0              0                     0

 Vlan    Dest MAC Failures   IP Validation Failures
 ----    -----------------   ----------------------
    1                    0                       0

Switch#
Switch#
Switch#
```



Expected: `Fa0/1` shows trusted, binding table shows PC0 MAC -> 192.168.10.10, drops counter increases on Fa0/2.

Demo 2: ARP Spoofing Blocked

1. Ensure PC0, PC1 have legit DHCP IPs.
2. From PC2 (Attacker) set static 192.168.10.20 and attempt to spoof gateway: `arp -a` and try to ping with fake ARP.
3. On switch:

```
show ip arp inspection vlan 1
show ip arp inspection statistics vlan 1
```

Expected log: `%SW_DAI-4-DHCP_SNOOPING_DENY: ... interface FastEthernet0/5` - Packet dropped because MAC/IP not in binding table.

7. Screenshots to Include in Report

1. Topology with green dots
2. `show ip dhcp snooping` - showing Fa0/1 trusted

3. `show ip dhcp snooping binding` - showing 2 bindings
4. `ipconfig /all` on PC0 - showing correct gateway 192.168.10.1
5. `show ip arp inspection statistics vlan 1` - showing Forwarded vs Dropped after attack

8. Conclusion

Without DHCP Snooping, any device can become a DHCP server and perform Man-in-the-Middle attacks. Without DAI, any device can poison ARP tables. By configuring one trusted port and enabling DAI with IP-MAC validation and rate limiting, we secured the LAN at Layer 2. This is a standard enterprise best practice for all access switches.

Best Practice: In real networks, also configure all uplink/trunk ports as trusted and enable `ip dhcp snooping` on all user VLANs.